

Q1. Obtain the principle conjunctive normal form.

$$(\sim p \Rightarrow r) \wedge (q \Leftrightarrow p)$$

Step 1: Simplify

$$\sim p \Rightarrow r \equiv p \vee r$$

$$q \Leftrightarrow p \equiv (q \wedge p) \vee (\sim q \wedge \sim p)$$

So,

$$E = (p \vee r) \wedge [(q \wedge p) \vee (\sim q \wedge \sim p)]$$

Step 2: Find rows where expression is false

false for:

$$(1, 0, 0), (1, 0, 1), (0, 0, 0), (0, 1, 0), (0, 1, 1)$$

Step 3: Write corresponding maxterms

PCNF:

$$(\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (p \vee q \vee r) \wedge (p \vee \sim q \vee r) \wedge (p \vee \sim q \vee \sim r)$$

Q2. Composite of function

$$f(x) = 2x + 1, \quad g(x) = x^2$$

(i) $g \circ f$

$$g(f(x)) = (2x + 1)^2 = 4x^2 + 4x + 1$$

(ii) $f \circ g$

$$f(g(x)) = 2x^2 + 1$$

Q3. Arrangement of Books

→ Subjects grouped together:

4 subject group → 4! ways

* Internal arrangements:

→ Math : $3!$

→ History : $4!$

→ Chemistry : $3!$

→ Biology : $2!$

Total ways:

$$4! \times 3! \times 4 \times 3! \times 2!$$

$$\Rightarrow 24 \times 6 \times 24 \times 6 \times 2$$

$$\Rightarrow 41472$$

Q4. Relation of perpendicular lines

→ Relation R:

LRm if line l is perpendicular to line m.

check properties

• Reflexive?

→ No line is perpendicular to itself → Not reflexive

• Symmetric?

→ if $l \perp m$, then $m \perp l$ → Symmetric

• Transitive?

→ if $l \perp m$ and $m \perp n$, then l is parallel, not perpendicular, to n → Not transitive

Conclusion

R is NOT an equivalence relation

Q5. ~~Set~~ Let $U = \{a, b, c, d, e, f, g\}$, $A = \{a, e, f\}$, $B = \{b, g\}$ and $C = \{d, e, g\}$ be sets, where U acts as the universal set. Determine following sets.

i) $A \times B$

ii) $(A \times B) - (A \times C)$

iii) $(A \times B^c) \cap (A \times C^c)$

→ Set operation

$$U = \{a, b, c, d, e, f, g\}$$

$$A = \{a, e, f\}, B = \{b, g\}, C = \{d, e, g\}$$

i) $A \times B$

$$\{(a, b), (a, g), (e, b), (e, g), (f, b), (f, g)\}$$

ii) $(A \times B) - (A \times C)$

$$A = \{a, e, f\}, B = \{b, g\}, C = \{d, e, g\}$$

So, $A \times B = \{(a, b), (a, g), (e, b), (e, g), (f, b), (f, g)\}$

$$A \times C = \{(a, d), (a, e), (a, g), (e, d), (e, e), (e, g), (f, d), (f, e), (f, g)\}$$

$$(A \times B) - (A \times C)$$

$$\rightarrow \{(a, g), (e, g), (f, g)\}$$

Remove $A \times B$

$$\text{Remaining pair: } \{(a, b), (e, b), (f, b)\}$$

iii) $(A \times B^c) \cap (A \times C^c)$

$$B^c = \{a, c, d, e, f\}, C^c = \{a, b, c, f\}$$

Common second element: $\{a, c, f\}$

Result:

$$\{(a, a), (a, c), (a, f), (e, a), (e, c), (e, f), (f, a), (f, c), (f, f)\}$$